Bad Apple!! In the Matrix

ECEN 106 – Computer Systems Final Project
Leslie Jury
December 13th,2022

Explanation: What my project does:

The internet trend of playing the famous music video for the song "Bad Apple!!", from the TouHou game series on every media platform you could imagine. Some of these platforms include everything from Desmos to a heart monitor. My project is inspired from this specific internet trend. My project is an LED Matrix showing the visuals of the Bad Apple music video on its 16x16 grid. Due to this ambitious project goal, I couldn't get the perfect result, however I have an advanced proof of concept in the hardware and software aspect of this project. I constructed a complex circuit to connect all four 8x8 matrix panels to each other. On this display I was able to display scrolling text of any text I entered. As for the programming aspect, I was able to convert each frame of the video to a thumbnail in hex code.

Explanation: How my project works:

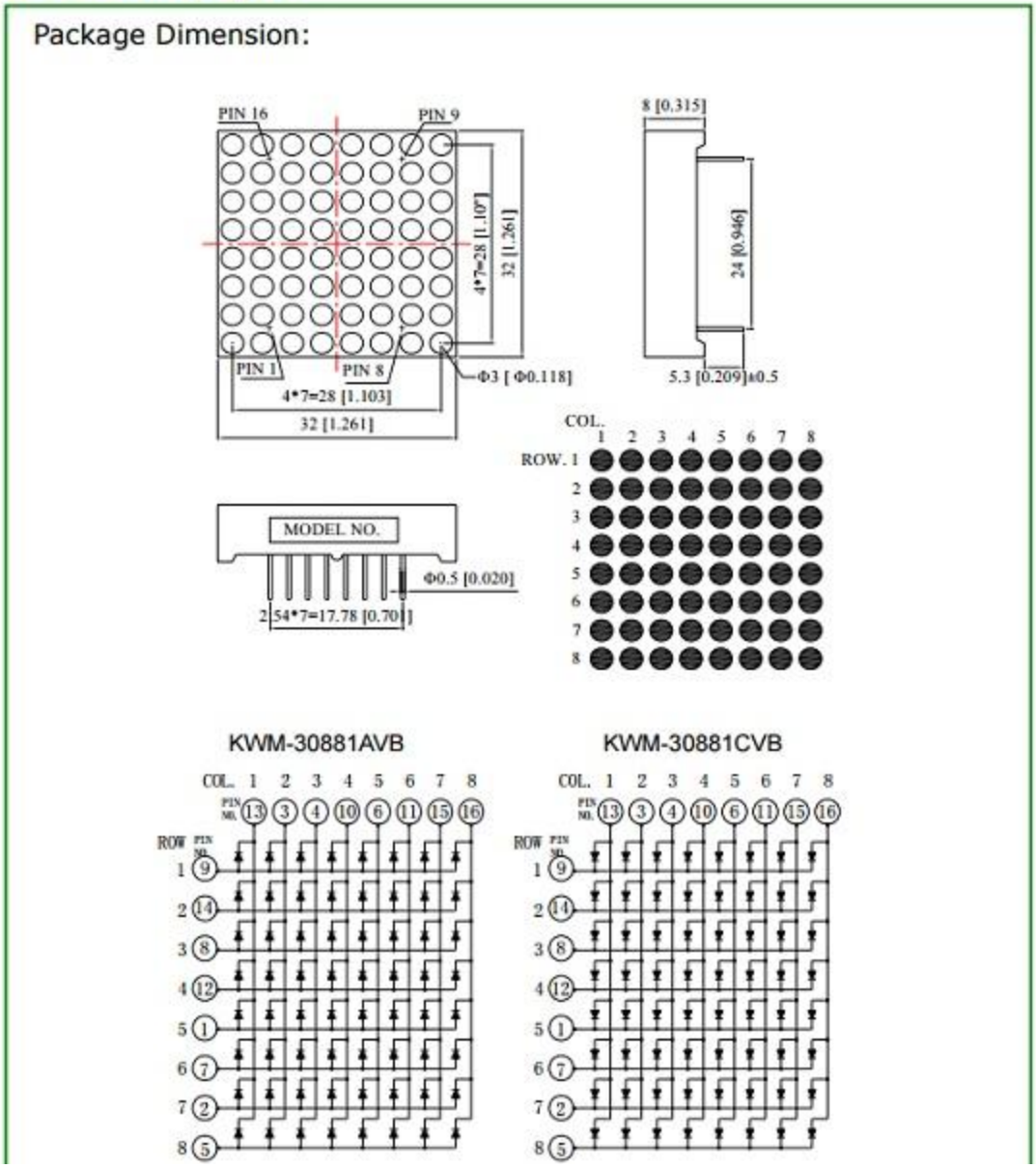
Software:

I used the Python Imaging Library and some written code to convert every frame into a thumbnail low resolution enough to display on a 16x16 grid. I then converted these thumbnail frames to hex-code to be inserted into the Arduino code I found.

Hardware:

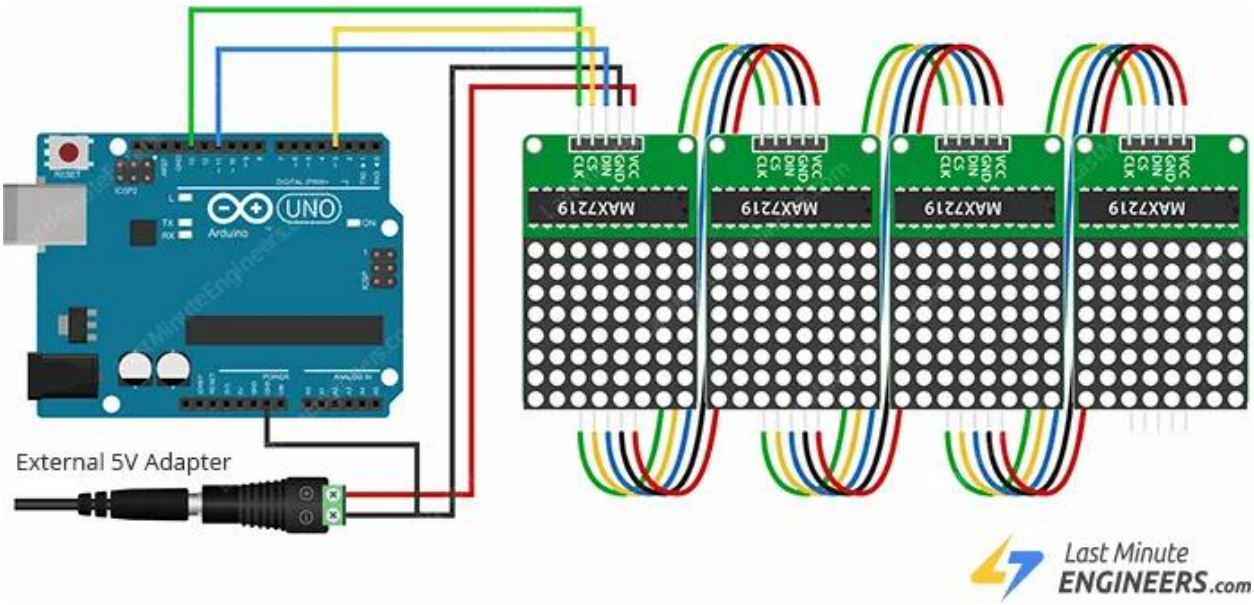
My wiring connects four 8x8 LED matrices to make them a 16x16 grid when put together. I wired them into the breadboard, then used jumper cables to connect each of the output ports to the next matrix. The power was sent to the first matrix, then through the connections of each of the output wires, power was distributed throughout.

Hardware: Wiring diagram



https://www.circuitspecialists.com/blog/getting-started-with-an-led-matrix-tutorial/

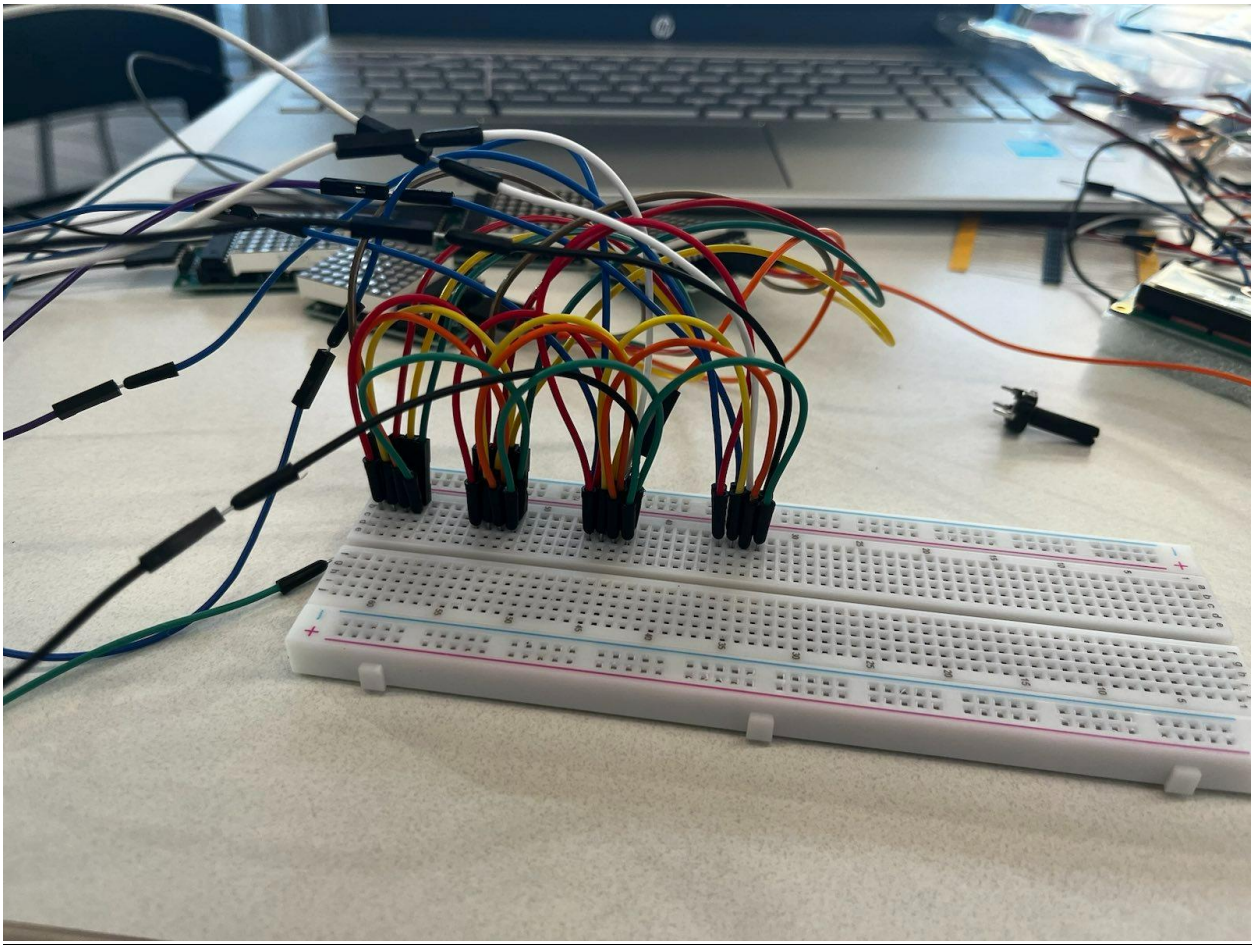
This is a diagram of the actual LED matrix, I looked at this to learn more about how matrices work.



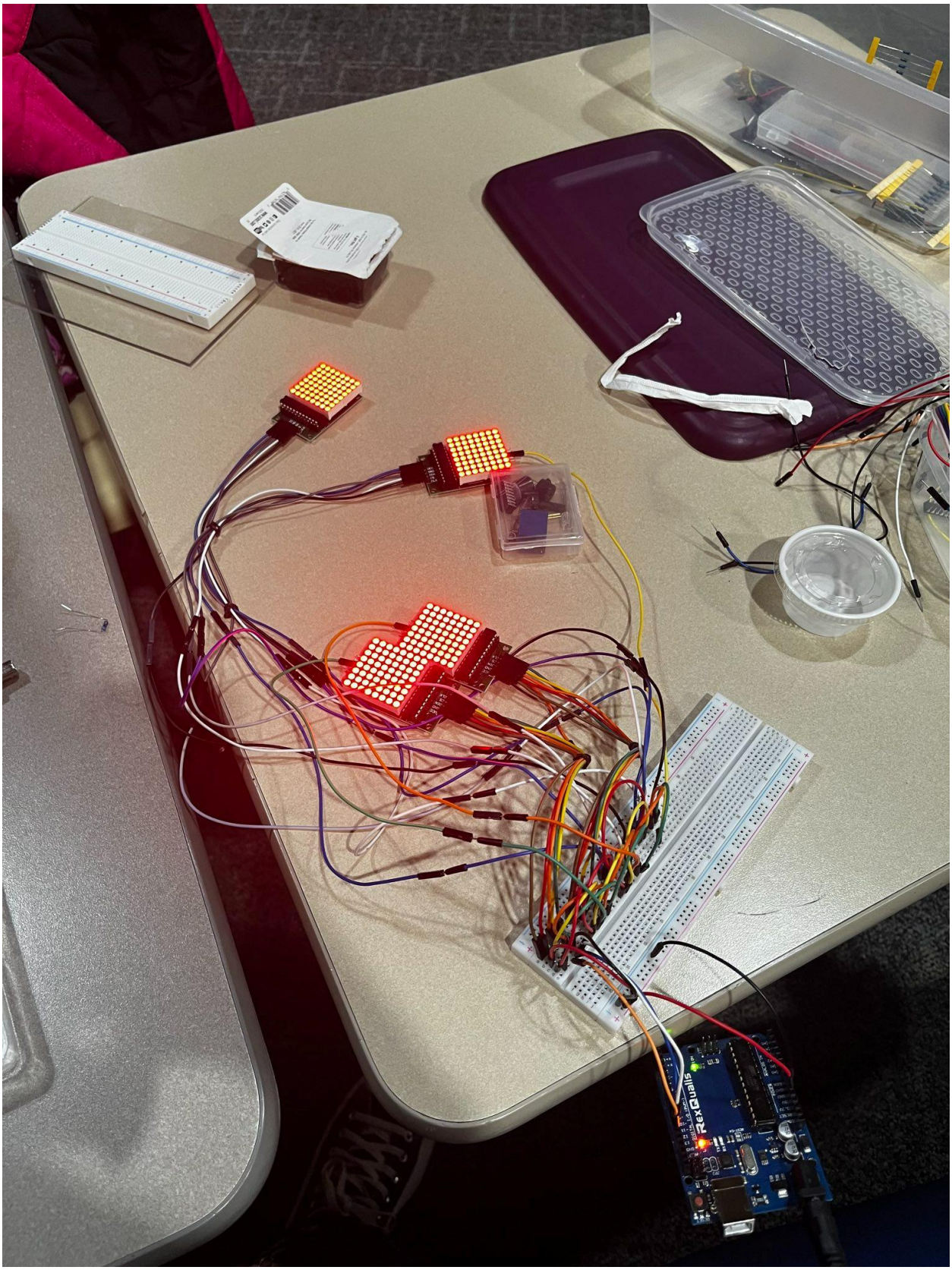
https://lastminuteengineers.com/

This diagram shows how to connect the matrices together, you can see the output chain I mentioned earlier in the explanation.

Hardware: Photo(s)



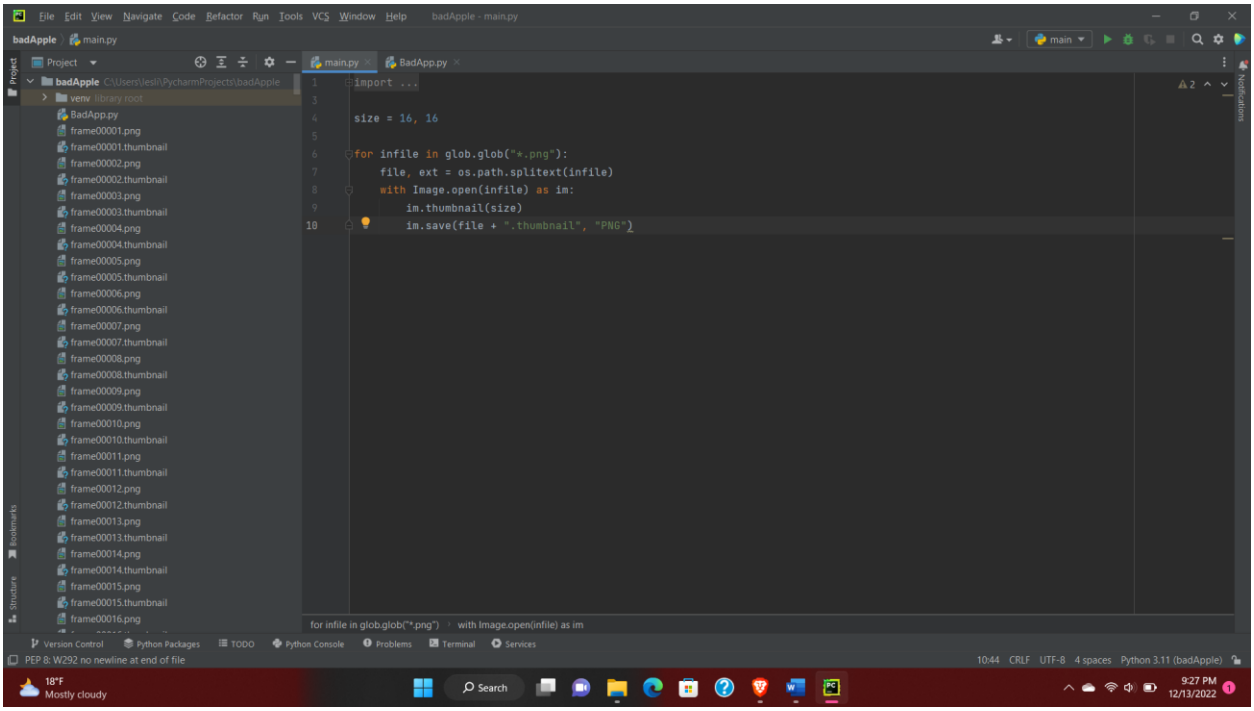
This is what my wiring looked like on my first day of wiring, you can see the four different sections on the board dedicated to each matrix.



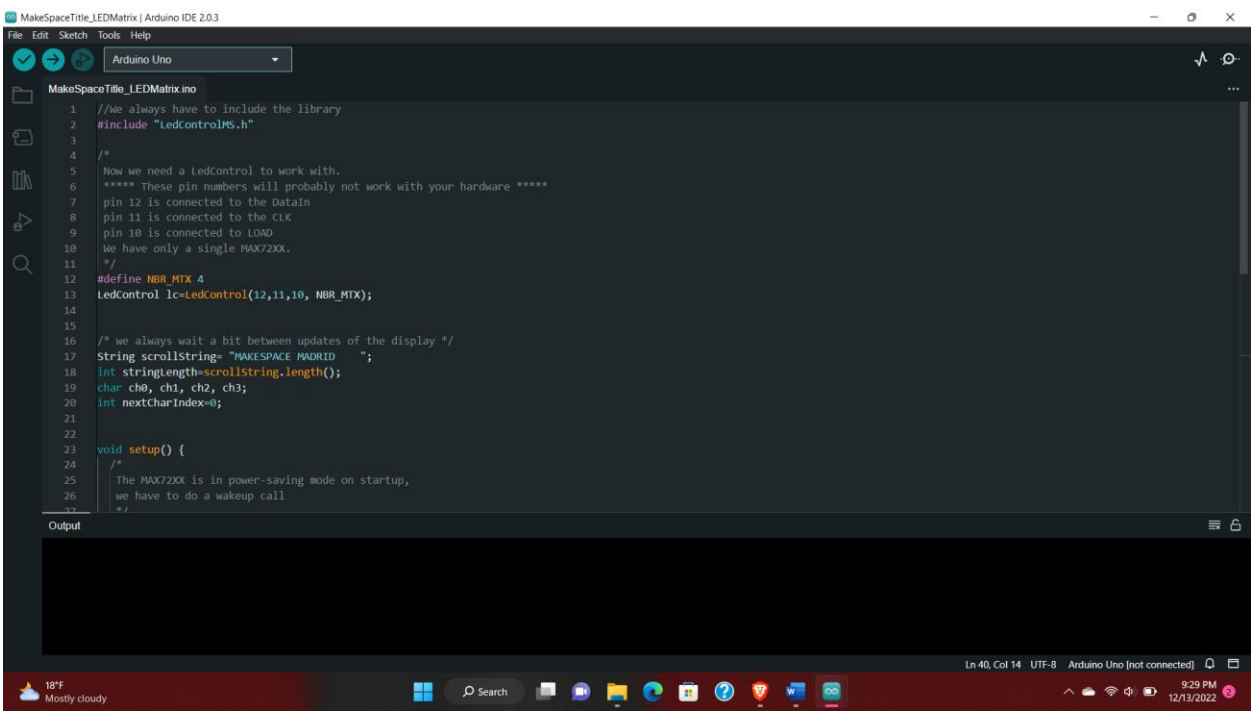
This is the wiring of my final project. You can see my cable management, and how I organized each

section of the board, I used a smaller section of the board to save space. It looks more clean and concise like this despite the added difficulty.

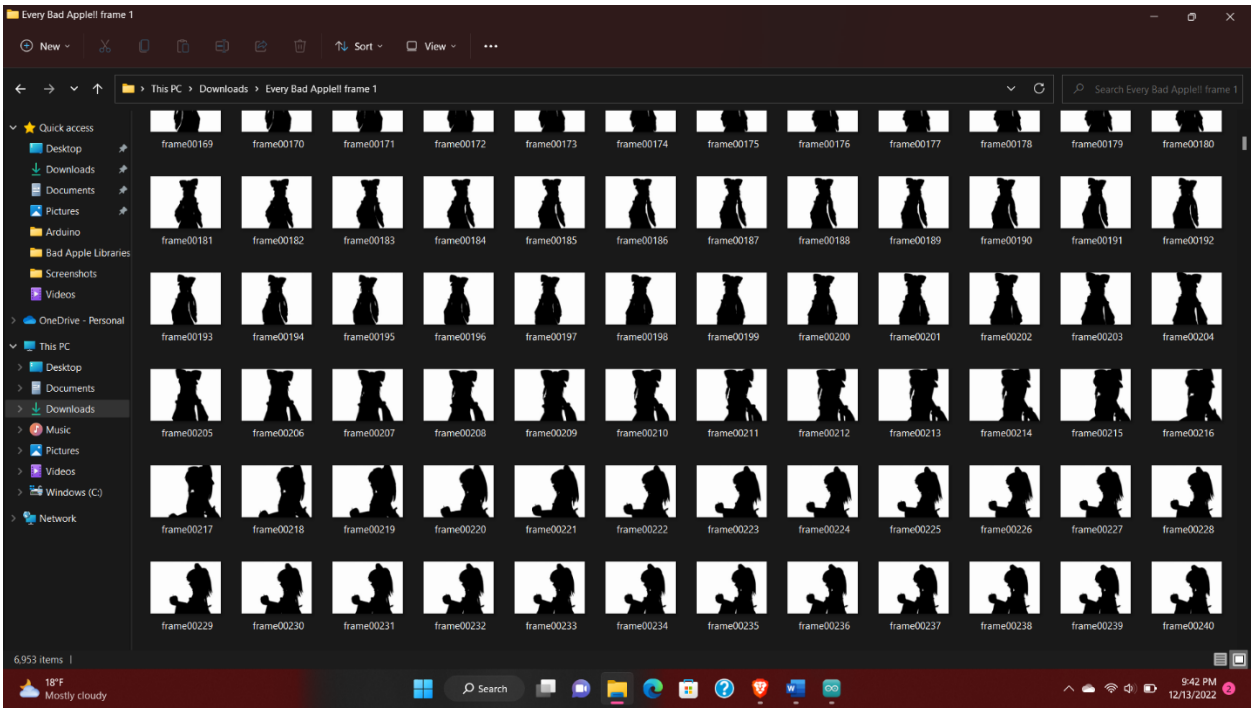
Software: Arduino sketch



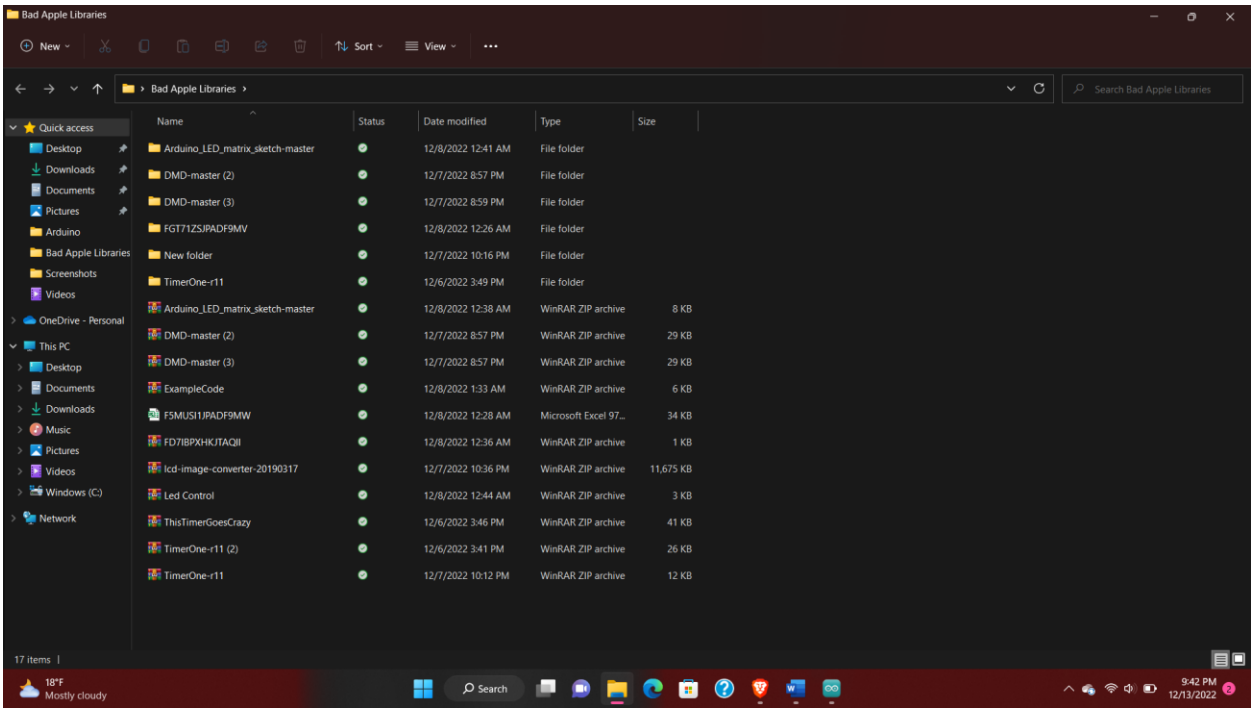
This is the code I used to turn each frame into a smaller pixel count.



This is the code I used in the Arduino IDE to display the text.



These are all of the frames from the music video. One unique aspect of the video is that it is only in black and white, this is the main reason why it is relatively easy to show on any platform because it has no complex lines or colors, just shapes. This was useful for making it hexcode, because it only needed on and off codes, no color.



These are all of my libraries I attempted to use, however there were too many errors for it to be worth the time.

Software: Citation

https://www.instructables.com/Multiple-LED-Matrixes-with-Arduino/
This instructable is the source I used as both my wiring guide and my demo material for the code.
I tried to customize it for this specific project, but the code didn't work. The only thing I changed was the text being displayed.

Discussion:

My biggest problem when making my final project was the Arduino libraries, I experimented with a lot of different code, however, most of them failed due to improper library downloading. I downloaded each library manually to fix this, however the programs were still facing errors related to the libraries. Luckily some of the code, specifically from my source website, had fewer library issues. I would say that is the only thing standing between my proof of concept and final draft. My experience with manually downloading libraries isn't quite on par with the necessary level.

My wiring was also an eventful journey, I carried a plastic bin with me everywhere the week I was finalizing everything, as the box would shake some, some wires would disconnect and I had to recall from memory their location, luckily after I did some cable management with hair elastics, this was much easier because each section was sorted well and was less confusing.

The programming also took a lot of trial and error. The project I used as inspiration was using serial communication, which I am not familiar with, and I do not have the proper hardware for this method. So instead, I decided to manually flip between frames and have all the hex code in the terminal of the IDE. This would have worked if the library issue wasn't present.